Intelligent Data Processing and Applications

COE 543/743

Spring 2026
Instructor: Joe Tekli

Project Proposal n#2

Wikipedia Semi-structured Infobox Document Clustering Tool

1. Title: Wikipedia Semi-structured Infobox Document Clustering Tool

2. Objective: The main objective of this project is to implement a semi-structured text document clustering tool, which can accept as input a collection of Wikipedia infobox documents, and then group similar documents together in coherent clusters, separating them from other less similar documents. In the context of this specific project, clustering will allow us to identify similar countries and group them into relevant clusters according to their similarities. For instance, Lebanon and Switzerland might be grouped together due to their high similarity (competed in Project 1), compared with France and Germany which could be part of another dedicated cluster.

3. Background and Motivations:

Clustering is the unsupervised classification of patterns (observations, data items, or feature vectors) into groups (clusters). The clustering problem has been addressed in many contexts and by researchers in many disciplines; this reflects its broad appeal and usefulness as one of the steps in exploratory data analysis. However, clustering is a difficult problem combinatorially, and differences in assumptions and contexts in different communities has made the transfer of useful generic concepts and methodologies slow to occur[1]. Clustering is useful in several exploratory pattern-analysis, grouping, decision-making, and machine-learning situations, including data mining, document retrieval, image segmentation, and pattern classification. However, in many such problems, there is little prior information (e.g., statistical models) available about the data, and the decision-maker must make as few assumptions about the data as possible. It is under these restrictions that clustering methodology is particularly appropriate for the exploration of interrelationships among the text document to make an assessment of their structure.

Grouping similar documents together can improve their storage and indexing[2], and thus positively affect the retrieval process. For instance, if two documents are similar, it is likely that they both either satisfy or not a given query. Therefore, when grouped together, similar documents would be much easier to retrieve than when scattered at different locations in the storage device. Clustering can also be critical in information extraction. It would be naturally much easier to identify and extract patterns from groups of similar text documents, rather than identifying/extracting patterns from a huge repository of heterogeneous documents. Current information extraction methods either implicitly or explicitly depend on clustering solutions[3].

5. Expected Results: The main objective of this project is to develop and implement two of the clustering algorithms covered in class. Each student group is expected to build on the foundations set in Project 1 (Wikipedia *Semi-structured Infobox Data Collection, Comparison, and Differencing*), using/refining their

[1] Algergawy A., Mesiti M., Nayak R., et al., *XML Data Clustering: An Overview.* ACM Computing Survey 2011. 43(4):25.
[2] Jain A.K.; Murty M.N. and Flynn P.J., *Data Clustering: A Review.* ACM Computing Surveys, 1999. 31(3):264-323.
[3] Tekli J., Chbeir R. and Yétongnon K., *An Overview of XML Similarity: Background, Current Trends and Future Directions.* Elsevier Computer Science Review, 2009. 3(3):151-173.

implemented document similarity measures and indices in order to develop the clustering tool. The tool should allow the user to set and fine-tune all parameters necessary to execute the implemented clustering algorithms. Also, the tool should present the clustering results in a meaningful way than can be easily understood by the user. Each group should implement two clustering algorithms (e.g., partitional, hierarchical divisive, hierarchical agglomerative, spectral, etc.).

Groups wishing to go the extra mile can include: cluster visualization functionality (e.g., visual representation of clusters, dendrograms, etc.), and cluster evaluation functionality (implementing cluster evaluation metrics to be evaluated with respect to user chosen clustering algorithms and reference data).

In the context of this specific project, clustering will allow us to identify similar countries and group them into relevant clusters according to their similarities. For instance, Lebanon and Switzerland might be grouped together due to their high similarity (competed in Project 1), compared with France and Germany which could be part of another dedicated cluster.

The choice of the programming environment (e.g., Java, C, C#, VB) is left to the students. Students can go the extra mile and use a document centric NoSQL database to effectively and efficiently store and access large numbers semi-structured documents. A **technical report**, of maximum 5 pages length, with proper documentation describing the implementation process, design choices and techniques exploited in developing the tool, is required per group of students. The report can include (but is not bound to) the following sections:

- Introduction: topic, objective(s) of the project, and potential applications,
- Background: context, problem analysis, and existing solutions
- Proposal: conceptual modeling, software application design and implementation,
- Experimental evaluation: test documents used, highlighting accuracy and execution speed of clustering solutions
- Conclusion: synthesis, personal experience, project perspectives in terms of possible extensions, improvements, and practical uses.

A **project presentation/demonstration** will also be conducted per group, of maximum 10 minutes duration. Presentations/demos will be organized at the **end of the semester** (to be confirmed). Project source codes and technical reports should be delivered to the instructor, in digital form, on the same day of the presentation, as google drive or drop box attachments sent to: jtekli@gmail.com (please do not use the instructor's LAU email which often filters out source codes and programming files).